

10-1 Volume with Whole Number Edges

Time Capsule Engineers | Standard: 6.G.2 | Color-by-Number Practice

Learning Goal: I can find the volume of a rectangular prism with whole-number edges using length $\times$ width $\times$ height.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: volume, length, width, height

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. Find the volume of a rectangular prism with $l=3$, $w=4$, $h=5$.

Work: _____

Answer: _____

3. Find the volume of a rectangular prism with $l=4$, $w=5$, $h=6$.

Work: _____

Answer: _____

5. Find the volume of a rectangular prism with $l=3$, $w=8$, $h=4$.

Work: _____

Answer: _____

7. Find the volume of a rectangular prism with $l=9$, $w=3$, $h=4$.

Work: _____

Answer: _____

9. Find the volume of a rectangular prism with $l=8$, $w=4$, $h=6$.

Work: _____

Answer: _____

11. Find the volume of a rectangular prism with $l=12$, $w=3$, $h=2$.

Work: _____

Answer: _____

2. Find the volume of a rectangular prism with $l=2$, $w=6$, $h=7$.

Work: _____

Answer: _____

4. Find the volume of a rectangular prism with $l=5$, $w=5$, $h=5$.

Work: _____

Answer: _____

6. Find the volume of a rectangular prism with $l=6$, $w=7$, $h=2$.

Work: _____

Answer: _____

8. Find the volume of a rectangular prism with $l=10$, $w=2$, $h=5$.

Work: _____

Answer: _____

10. Find the volume of a rectangular prism with $l=7$, $w=5$, $h=3$.

Work: _____

Answer: _____

12. Find the volume of a rectangular prism with $l=11$, $w=4$, $h=5$.

Work: _____

Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

10-1 Mystery Picture: Engineering Tile Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
60 = red	84 = orange	120 = yellow	125 = green
96 = blue	108 = purple	100 = pink	192 = brown
105 = gray	72 = teal	220 = black	

2	2	1	2	2	2	3	3	3	4	4	6
2	5	5	5	5	5	6	6	6	7	9	6
1	5	8	9	8	8	9	9	9	12	9	6
3	6	10	2	2	1	2	2	4	2	11	6
3	6	9	2	5	5	5	7	4	2	1	6
3	6	9	1	5	8	11	7	4	12	9	6
5	8	11	3	6	12	1	10	4	2	11	8
5	8	9	3	4	8	10	8	5	2	11	8
5	8	11	7	5	5	4	5	5	3	11	8
7	10	3	4	4	2	5	3	4	5	2	10
7	12	12	1	6	11	12	12	12	2	2	11
9	9	9	8	8	8	9	9	9	2	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	60	red	$V = \text{length} \times \text{width} \times \text{height}$.
2	84	orange	$V = \text{length} \times \text{width} \times \text{height}$.
3	120	yellow	$V = \text{length} \times \text{width} \times \text{height}$.
4	125	green	$V = \text{length} \times \text{width} \times \text{height}$.
5	96	blue	$V = \text{length} \times \text{width} \times \text{height}$.
6	84	orange	$V = \text{length} \times \text{width} \times \text{height}$.
7	108	purple	$V = \text{length} \times \text{width} \times \text{height}$.
8	100	pink	$V = \text{length} \times \text{width} \times \text{height}$.
9	192	brown	$V = \text{length} \times \text{width} \times \text{height}$.
10	105	gray	$V = \text{length} \times \text{width} \times \text{height}$.
11	72	teal	$V = \text{length} \times \text{width} \times \text{height}$.
12	220	black	$V = \text{length} \times \text{width} \times \text{height}$.

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.